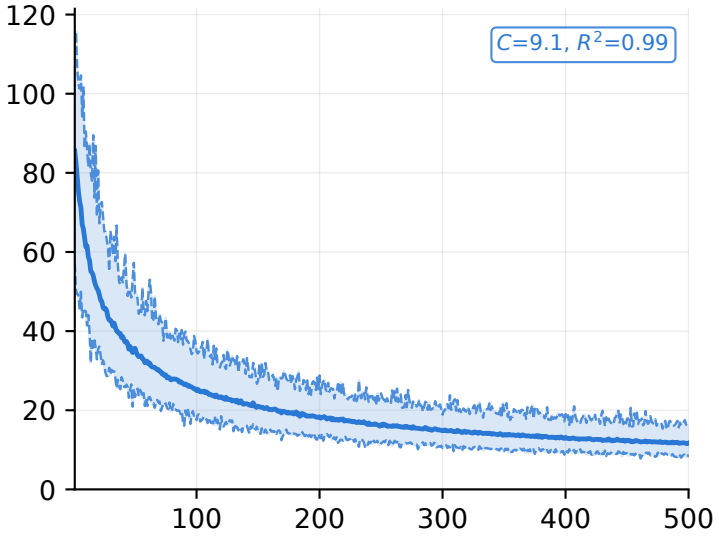
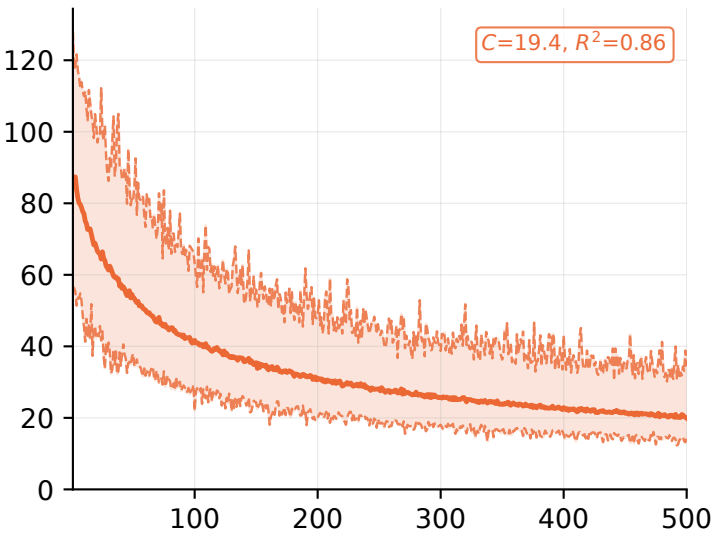
DiM direction split-half angle vs. subsample size, across models and datasets
(solid: mean; shaded/dashed: min–max over 200 draws; fit: $\tan^2(\theta_{split}/\sqrt{2}) \approx C/n$)

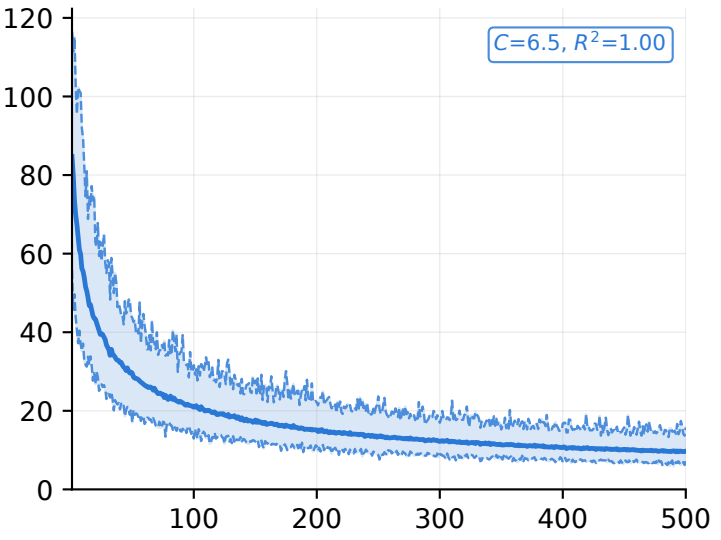
TruthfulQA — llama3-8B-Instruct



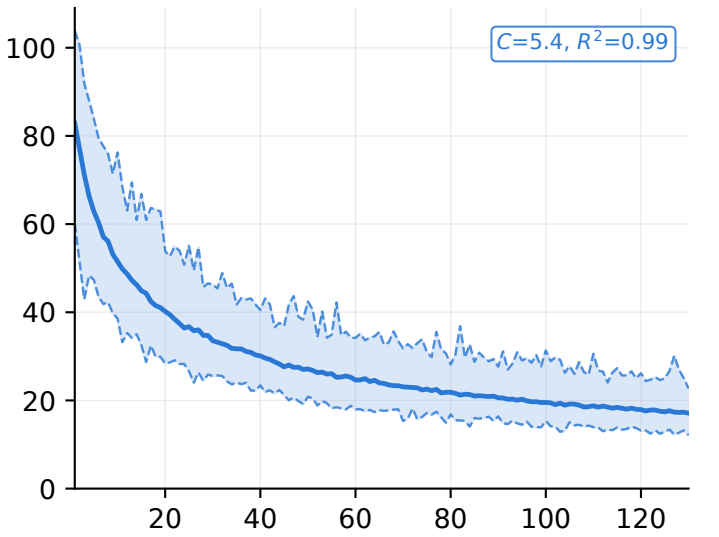
TruthfulQA — qwen2.5-7B-Instruct ⚠



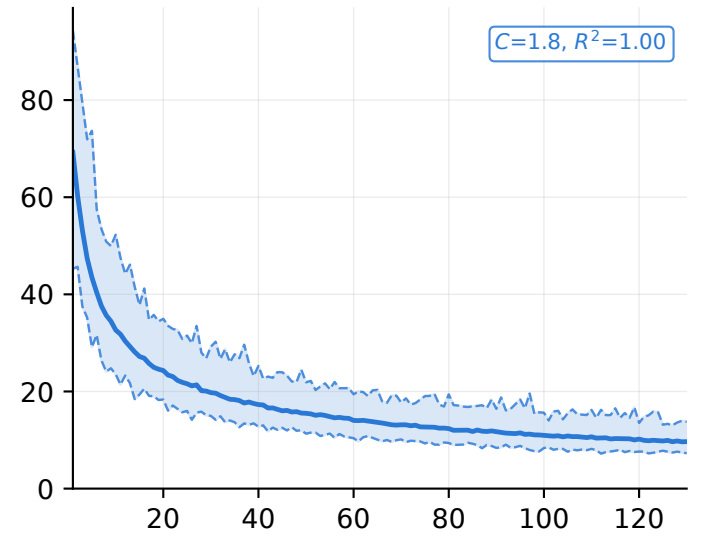
TruthfulQA — qwen3-8B*



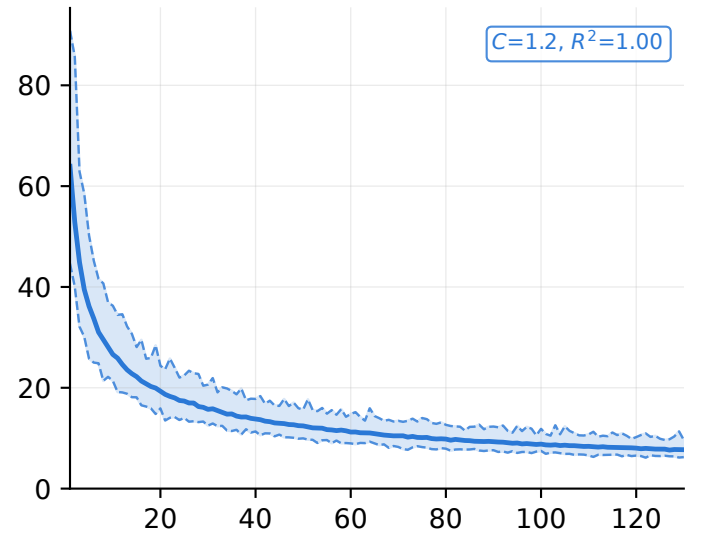
Refusal — llama3-8B-Instruct



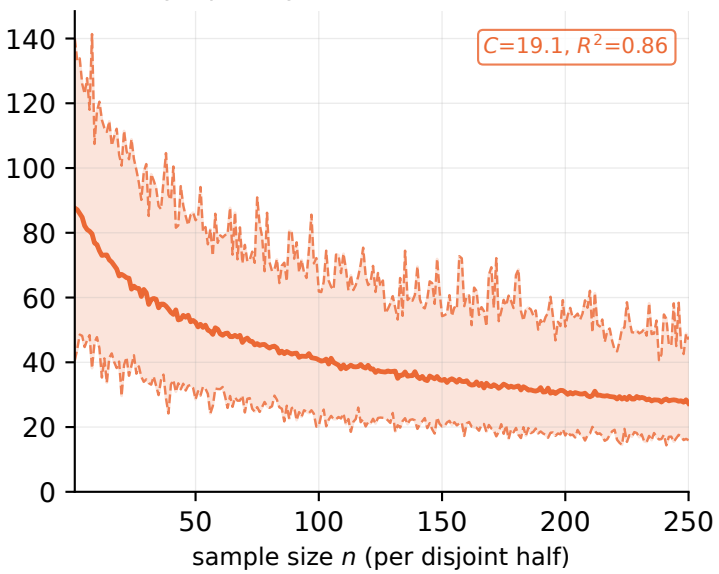
Refusal — qwen2.5-7B-Instruct



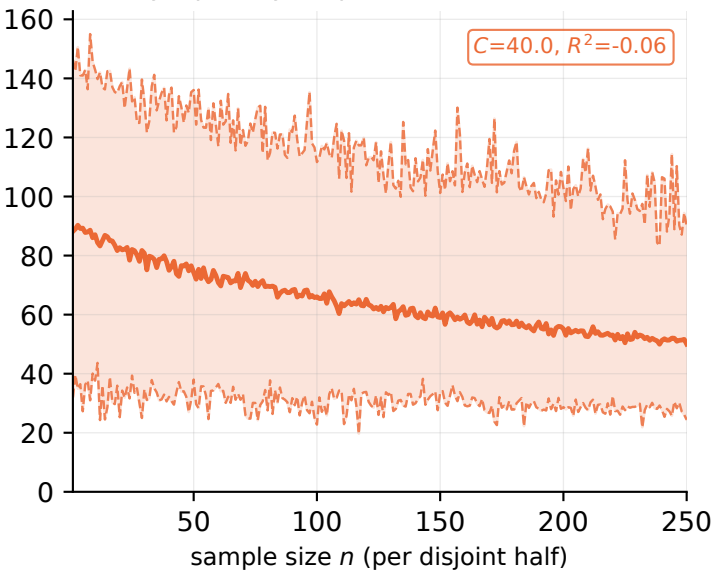
Refusal — gemma-2-9B-it



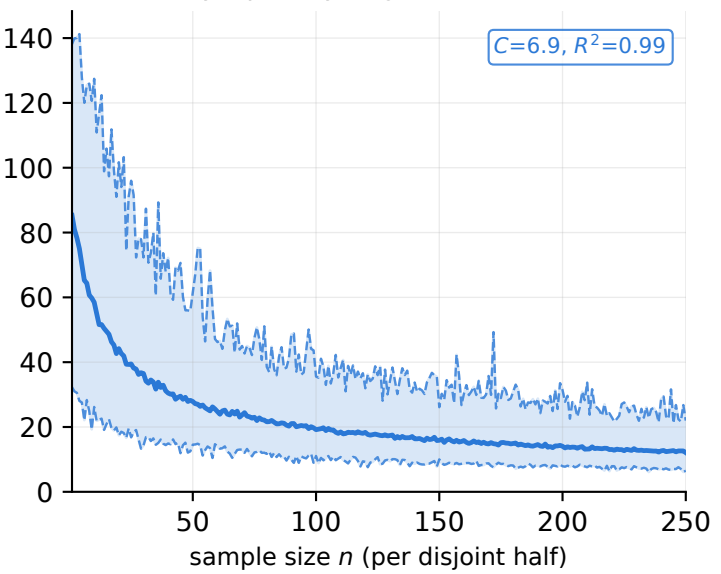
Sycophancy — llama3-8B-Instruct ⚠



Sycophancy — qwen2.5-7B-Instruct ⚠



Sycophancy — gemma-2-9B-it



*TruthfulQA column 3 shows qwen3-8B (no gemma-2-9B-it TruthfulQA run). Orange panels (⚠, R^2<0.9) show a poor 1/n fit — a signal of pool heterogeneity (e.g. subpopulations whose DiM directions disagree), not just sampling noise.